Name:_____

Exam Question Booklets
OCR Physics A

Booklet 9
14 Questions
96 marks

Things to remember:

✓

✓

✓

✓

✓

✓

✓

1

(a) Define

(i) stress .....

.....

(ii) strain .....

.....[2]

(b) **(i)** Distinguish between elastic and plastic behaviour when materials are stretched.

.....

.....

.....[2]

(ii) Define elastic limit.

.....

.....[2]

(c) **(i)** State the SI unit of the Young modulus.

unit ..... [1]

(ii) Describe, with the aid of a diagram, an experiment to determine the Young modulus of steel in the form of a wire. Explain how to use your readings to obtain the Young modulus. (You will be awarded marks for the quality of written communication in this part of the question.)

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..... [7]

2 This question is about a nuclear fusion reaction.

(a) Fig. 1.1 shows a deuterium and a tritium nucleus travelling directly towards one another. At a large distance apart, each incident nucleus is travelling at a speed v. When they collide and fuse an alpha particle and a neutron are formed. They travel apart as shown. The separation speeds of the alpha particle and neutron are shown in the figure. The mass of each nucleus, in terms of m, the mass of a nucleon, is given in the centre of each particle.

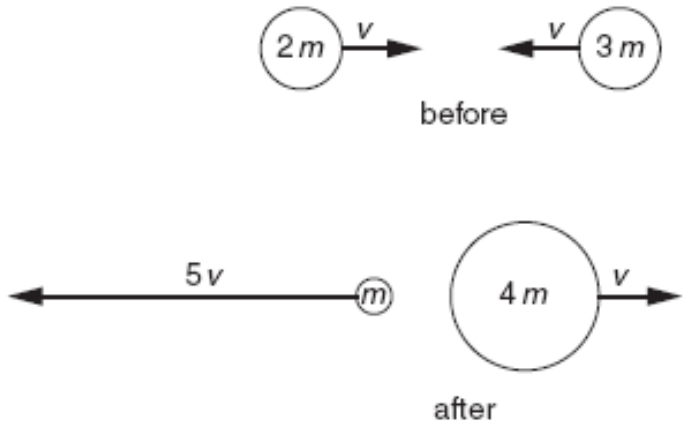

Fig. 1.1

(I) Show that momentum is conserved in the interaction. Work in terms of m and v.

[2]

(II) Show that the total kinetic energy of the particles increases by $12mv^2$ during the fusion reaction.

[2]

(III) Compare the increase in total kinetic energy of the particles with the kinetic energy of the neutron and comment.

[2]

(iv) Calculate the increase in total kinetic energy in J of the particles using the data below.

Data:
$m = 1.67 \times 10^{-27}$ kg
$v = 1.19 \times 10^{7}$ m s^{-1}

increase = ... J **[1]**

3 A cricketer throws a cricket ball of mass 0.16 kg.

(a) Fig. 1.1 shows how the force on the ball from the cricketer's hand varies with time. The ball starts from rest and is thrown horizontally.

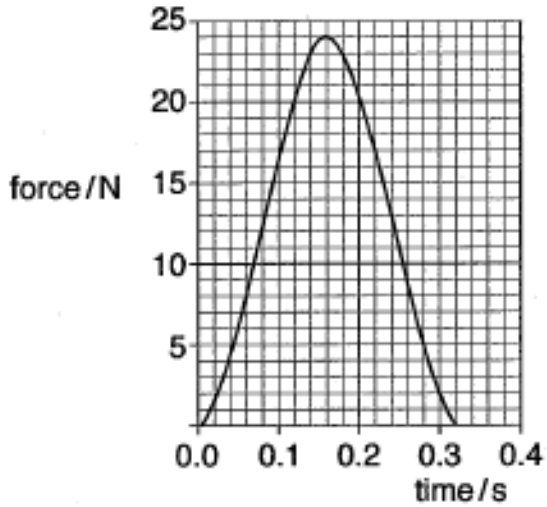

Fig. 1.1

(i) Estimate the area under the graph.

area = ... N s [1]

(ii) The area under the graph represents a change in a physical quantity for the ball. State the name of this quantity.

.....[1]

(iii) Calculate the speed of the ball, mass 0.16 kg, when it is released.

speed = ... $m\,s^{-1}$ [2]

(iv) Calculate the **maximum** horizontal acceleration of the ball.

acceleration = ... $m\,s^{-2}$ [2]

A table-tennis ball is released from rest to fall to the floor. Air resistance causes the ball to reach terminal velocity before it reaches the floor. Fig. 1.1 shows graphs of the potential and kinetic energies of the ball against the distance fallen. Only the energy axis has been given a scale.

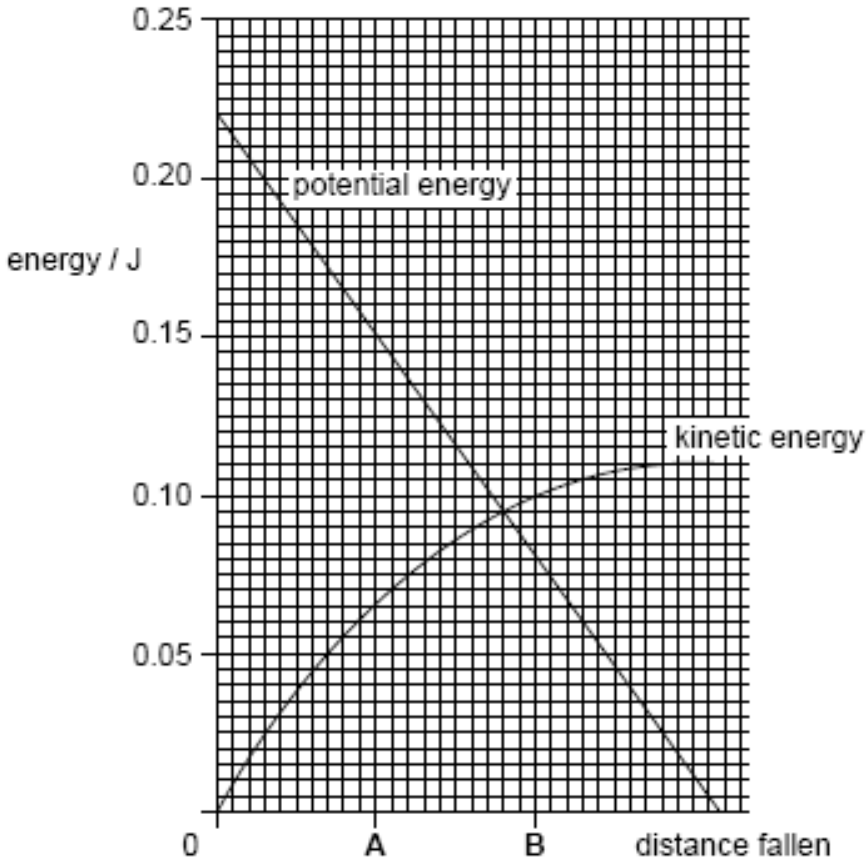

Fig. 1.1

The mass of the ball is 0.014 kg.

(a) Use Fig. 1.1 to calculate the height *h* from which the ball is dropped. The floor is taken as the zero of potential energy.

h = ... m [2]

(b) Describe how the kinetic energy of the ball is changing as it falls between **A** and **B** as indicated on Fig. 1.1.

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.....

.....[2]

(c) **(i)** Show that the terminal velocity of the ball is about $4\,m\,s^{-1}$.

[2]

(ii) Calculate the drag force on the ball at its terminal velocity.

drag force = ... N [1]

(iii) Calculate the energy loss per second as the ball falls at its terminal velocity.

energy loss per second = ... W [2]

(d) The ball bounces on the floor. 20% of the ball's kinetic energy is lost in the bounce. Calculate the change in momentum of the ball at the bounce. Give a suitable unit for your answer.

momentum change = ... unit..... [4]

5

(i) State the SI base units of volume.

base units of volume ..... [1]

(ii) Show that the SI base units of pressure are $kg\,m^{-1}\,s^{-2}$.

[1]

6

The speed, frequency and wavelength of the light beam in air are v, f and λ respectively.

(i) By placing ticks (✓) in the appropriate boxes of Fig. 1.2 show which of these terms increase, decrease or stay the same when the light enters liquid. [3]

	increase	decrease	stay the same
speed			
frequency			
wavelength			

(a) Draw a labelled diagram to show how plane water waves, in a ripple tank, are diffracted as they pass through a narrow rectangular slit.

[3]

(b) Fig. 4.1 shows a piano being played on one side of a wall. Suggest why the person standing on the other side of the wall can hear the piano clearly even though she cannot directly see it.

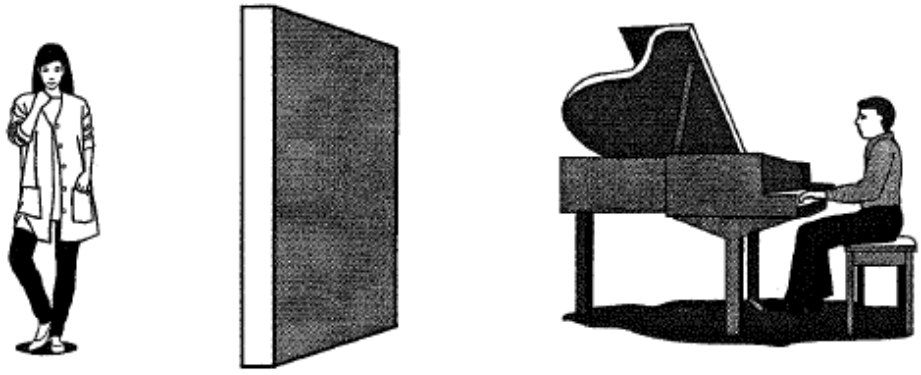

Fig. 4.1

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..... [2]

8 **(a)** Fig. 3.1 shows, at a given instant, the shape of a stretched rope along which a transverse wave is travelling from left to right. The letters V, W, X, Y and Z identify five particles on the rope.

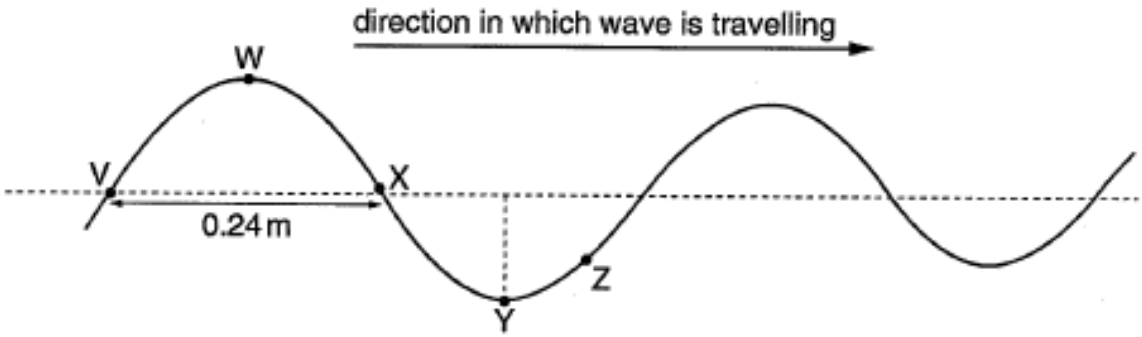

Fig. 3.1

(i) On Fig. 3.1 sketch the shape of the rope a short time later. [1]

(ii) On Fig. 3.1 draw arrows to show the directions in which the particles X and Z are moving during this short time. [2]

(iii) The distance between V and X is 0.24 m (as shown). The frequency of the wave is 3.6 Hz. Calculate the speed of the wave.

wave speed = ... $m\,s^{-1}$ [3]

(iv) The frequency is doubled to 7.2 Hz. Determine the changes, if any, in

1. the wave speed

.....

.....

2. the wavelength.

.....

..... [3]

(a) **(i)** State what is meant by a *progressive* wave.

.....

..... [1]

(ii) Describe in terms of the motions involved, the essential difference between longitudinal and transverse progressive waves.

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[2]

(b) Describe how

- plane, transverse water waves can be produced in a ripple tank
- the wavelength of these waves could be increased
- the speed of these waves could be reduced.

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..... [3]

(a) Describe the motion of free electrons

(i) in a metal

.....

.....

..... **[3]**

(ii) in a metal wire carrying an electric current.

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.....

.....

..... **[3]**

(b) A metal wire of cross-sectional area A contains n free electrons per m^3. When the wire carries a current I, the drift velocity of free electrons in the wire is v. The electronic charge is e.

(i) Explain what is meant by *drift velocity*.

.....

..... **[1]**

(ii) Show that the current in the wire is given by the expression $I = nAve$.

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..... **[3]**

(a) Fig. 1.1 shows an electrical circuit.

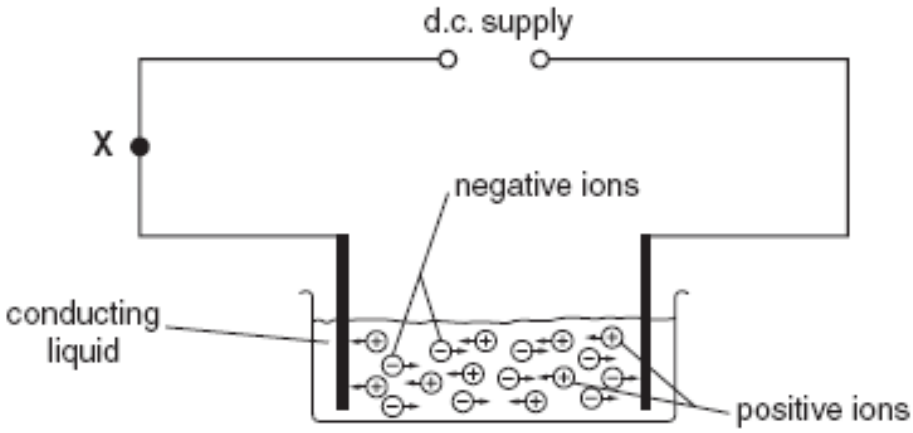

Fig. 1.1

(i) The directions of flow of ions in the liquid are shown. On Fig. 1.1, draw an arrow at **X** to show the direction of the electron flow in the wire. [1]

(ii) State what is meant by *conventional current*.

.....

.....[1]

(iii) A charge of 0.76 C flows past point **X** in a time of 5.0 minutes. Calculate the current in the wire.

current = ... A [3]

(a) Name one common property of electromagnetic waves not shared by other waves.

..... **[1]**

(b) Fig. 5.1 shows a block diagram of the seven regions of the electromagnetic spectrum, labelled **A** to **G**.

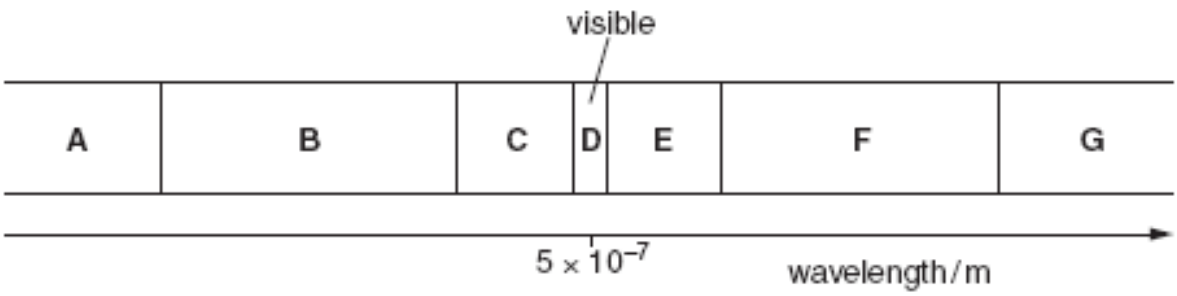

Fig. 5.1

Name the principal radiation in each of the regions **A**, **C** and **F**.

A ..... **C** .....

F ..... **[3]**

Fig. 5.1 represents waves on the surface of water in a ripple tank at one particular instant of time.

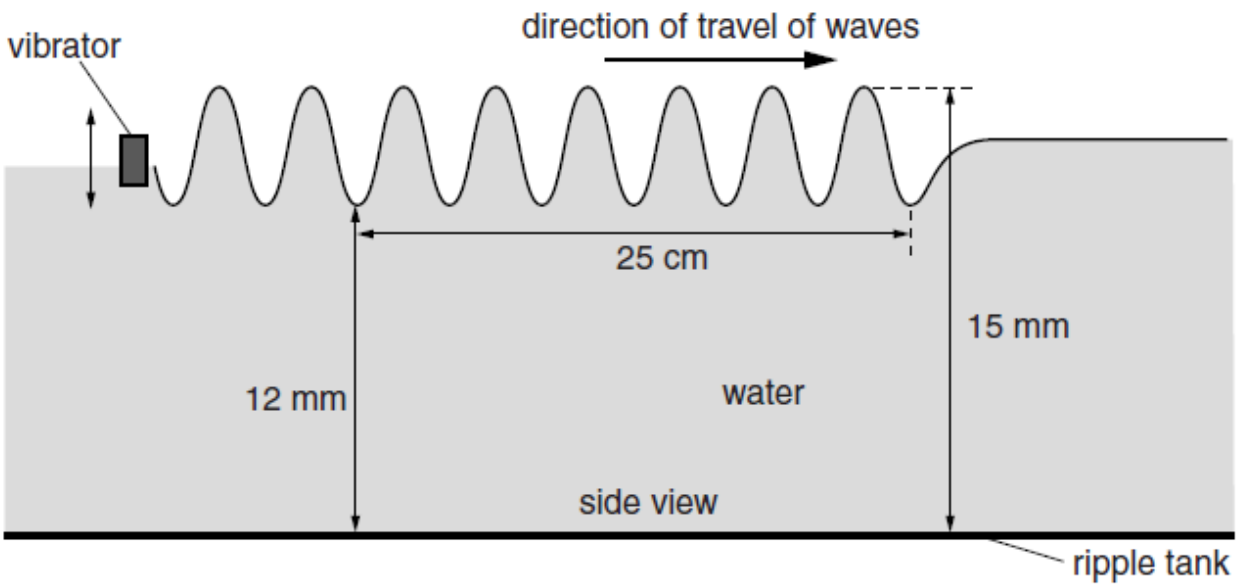

Fig. 5.1 (not to scale)

A vibrator moves the surface of the water to produce the waves of frequency *f*. The speed of the waves is 7.5 cm s^{-1}. Where the waves travel on the water surface, the maximum depth of the water is 15 mm and the minimum depth is 12 mm.

(i) Calculate, for the waves,

1. the amplitude,

amplitude = ... mm [1]

2. the wavelength.

wavelength = ... m [2]

(ii) Calculate the time period of the oscillations of the vibrator.

time period = ... s [2]

(a) Describe the differences between transverse and longitudinal waves.

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.....[2]

(b) Fig. 3.1 shows a **progressive** longitudinal wave formed in a slinky spring by an oscillator connected to a signal generator.

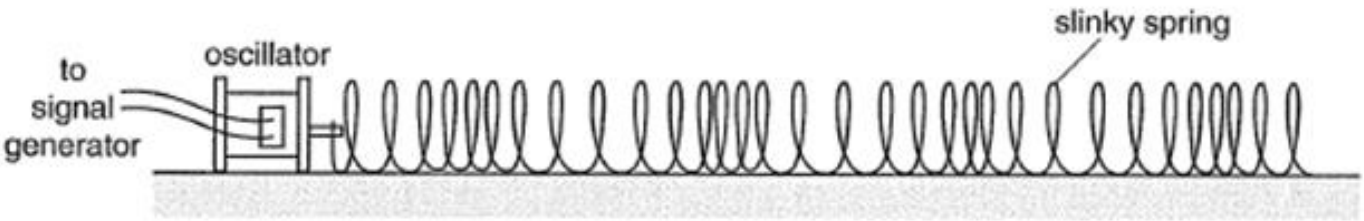

Fig. 3.1

On Fig. 3.1,

(i) draw arrows to show the direction of the vibrations produced by the oscillator – label these **V** [1]

(ii) label with a **C** the centre of a compression on the slinky [1]

(iii) show the wavelength of the wave and label this λ. [1]

(c) State and explain the effect on the wavelength of increasing the frequency of the oscillator.

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.....[2]